Rhizotomos.

Which of Dioscorides' 2,000 year old remedies has modern science actually tested, and which did it never test?

Live atlas · **rhizotomos.vercel.app**

Code · **github.com/Arnav020/rhizotomos**

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ῥιζοτόμος: "root cutter", the herb gatherers Dioscorides learned from

"Ancient remedies validated by science."
Which ones?

De Materia Medica (c. 60 CE) was the pharmacology reference of Europe, Byzantium and the Islamic world for 1,500 years. Nobody can say which of its claims modern research has tested.

OBSTACLE 1

Names drift

Greek **thymon** is not garden thyme. "White hellebore" is **Veratrum**. **Silphion** is extinct.

OBSTACLE 2

Popularity bias

Garlic appears in thousands of papers, so raw counts make every claim about it look "supported".

OBSTACLE 3

Evidence is not evidence

A hundred in vitro papers are not one clinical trial. Some evidence hides inside isolated compounds.

One ancient text, two open APIs, one regulatory check

34

curated claims from *De Materia Medica*: Greek name, candidate taxon, paraphrased use, name certainty, reading certainty, curator note

204

PubMed E-utilities queries: records, clinical trials, systematic reviews, taxon totals and 68 negative controls

34

taxa resolved on the GBIF Backbone with occurrence records worldwide and in Greece (CC BY 4.0)

11

claims overlapping a current EMA/HMPC herbal monograph, held out as an independent check

Every API response is cached in the repository, so the whole analysis reruns offline. Counts only: no abstracts are stored.

Sources: Dioscorides (public domain) · GBIF.org · NCBI PubMed · EMA/HMPC

From a Greek plant name to a research priority

01 CLAIM

Read

Greek name, use and two doubt flags

→

02 TAXON

Resolve

GBIF accepted name, confidence, occurrences

→

03 EVIDENCE

Weigh

PubMed records, trials, reviews ÷ taxon total

→

04 GAP

Score

signal × (1 − trial saturation) × certainty

Lift over own controls

Each plant is compared with itself paired with 2 unrelated uses. Popularity cancels out.

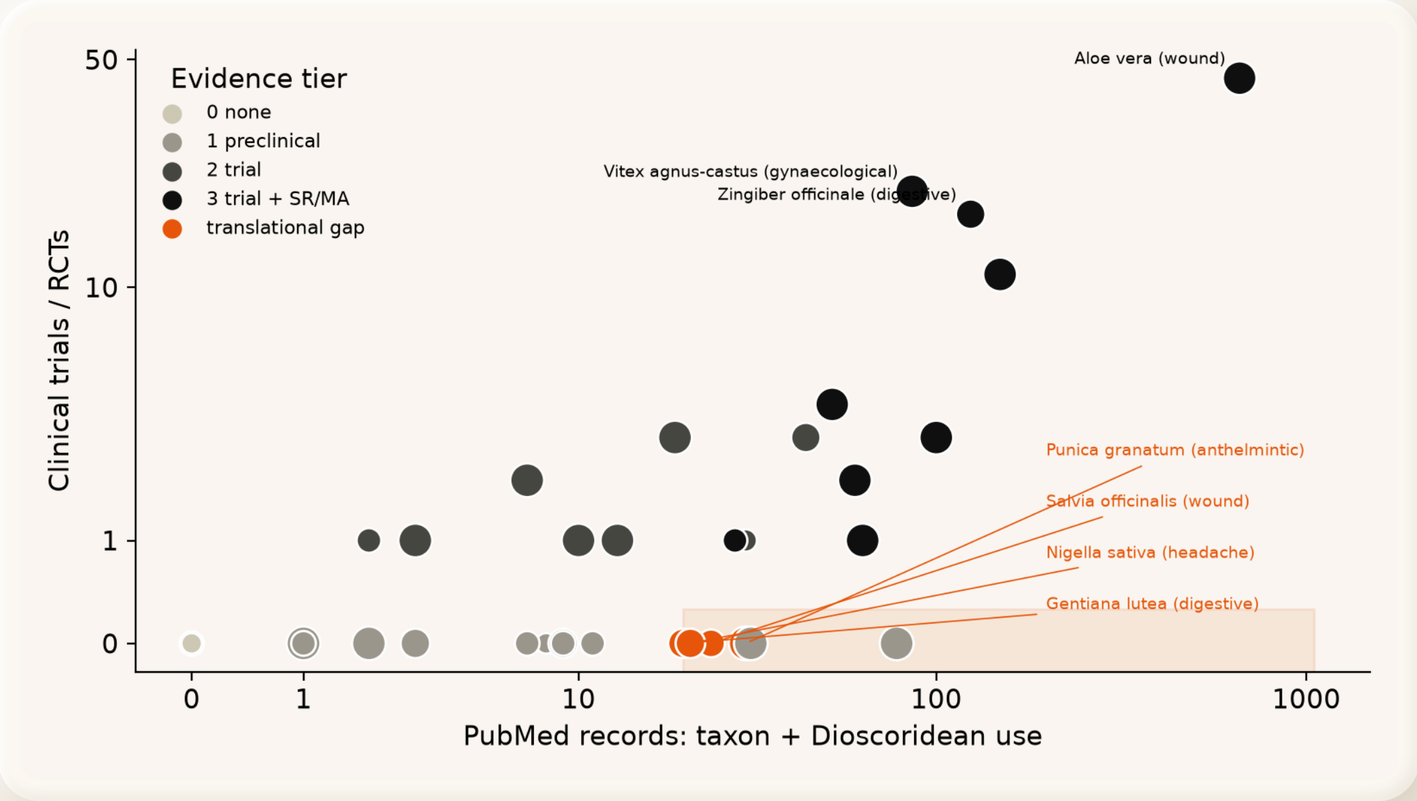
Evidence tier 0 to 3

none · preclinical only · ≥1 clinical trial · trial plus systematic review

Doubt is data

Uncertain identifications are down-weighted, never hidden.

Only 17 of 34 claims ever reached a human trial



4

translational gaps: 20+ papers, zero clinical trials

3

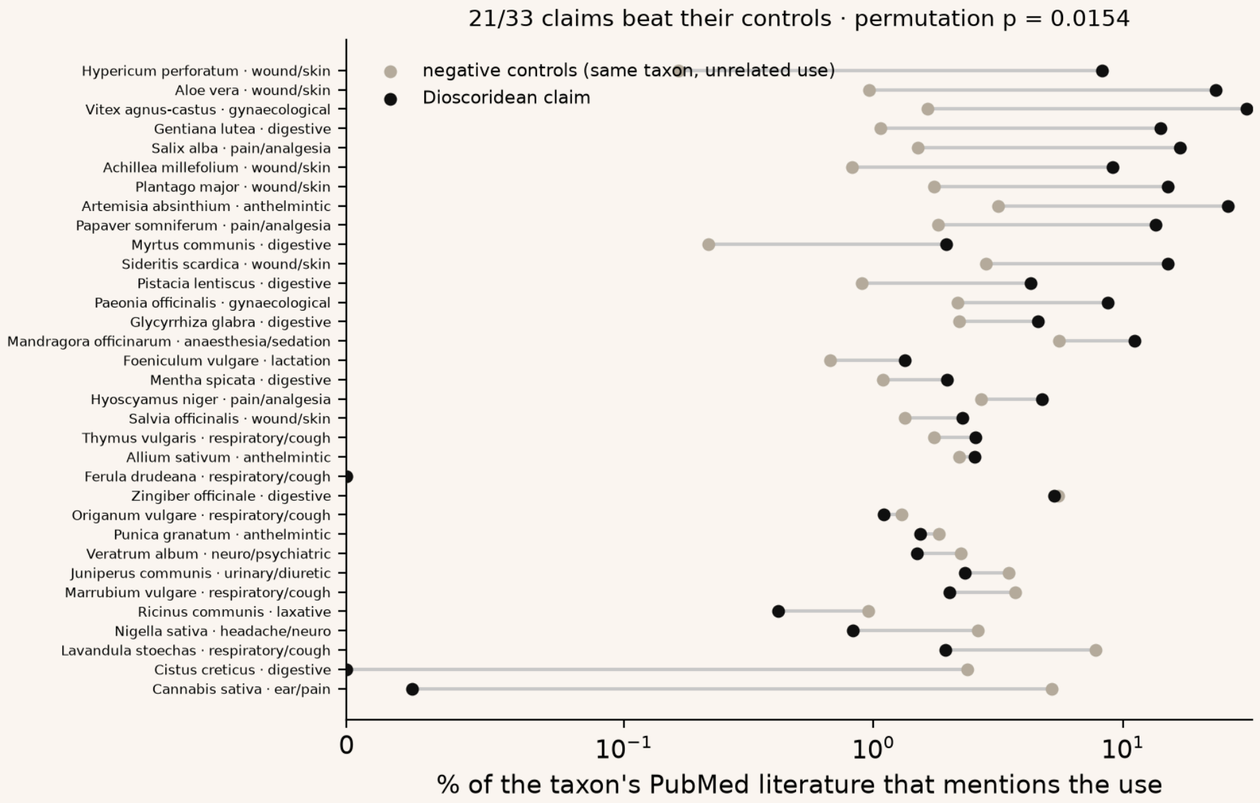
translated via an isolated compound, the "Papaver paradox"

2

claims with no modern literature at all, including lost silphion

PubMed counts retrieved 9 Nov 2026

The signal is real, not an artefact of famous plants



21/33

claims beat their own negative controls · median lift 1.7× · Wilcoxon p 0.011 · permutation p 0.015

AUC 0.82

claims in EU herbal monographs sit in higher evidence tiers · $p < 0.001$

p 0.019

still significant without low certainty identifications

Sign flip permutation, 20,000 draws, seed 42

Where 2,000 years of use never met a clinical trial

DIOSCORIDES, BOOK III

γεντιανή

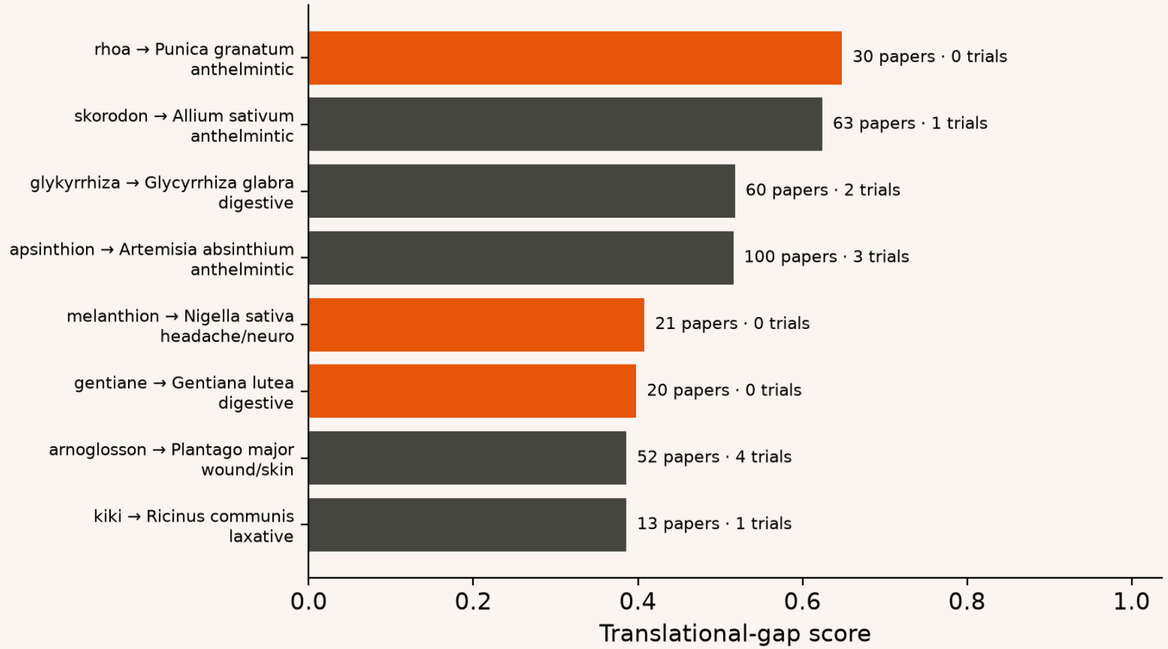
Gentiana lutea · gentian root

"warms, and is good for the stomach"

12× lift over controls

20 papers

0 trials



Hypotheses for trial prioritisation, not medical claims. Toxicity and the quality of preclinical work must be assessed first.

Databases are confident. History is not.

THE TAXONOMY BLIND SPOT

99 to 100%

GBIF match confidence for all 34 names, including disputed ones like thymon, sideritis and silphion. Modern taxonomy cannot see historical identification error, so Rhizotomos stores it as a certainty weight.

THE PAPAVER PARADOX

0 trials

for *Papaver somniferum* and pain, although opioids are the most successful plant drug in history. Evidence migrates into compounds (morphine, scopolamine, aspirin) and plant name searches miss it. We label these claims instead of calling them gaps.

MEKON

Opium poppy → morphine

MANDRAGORAS

Mandrake → scopolamine

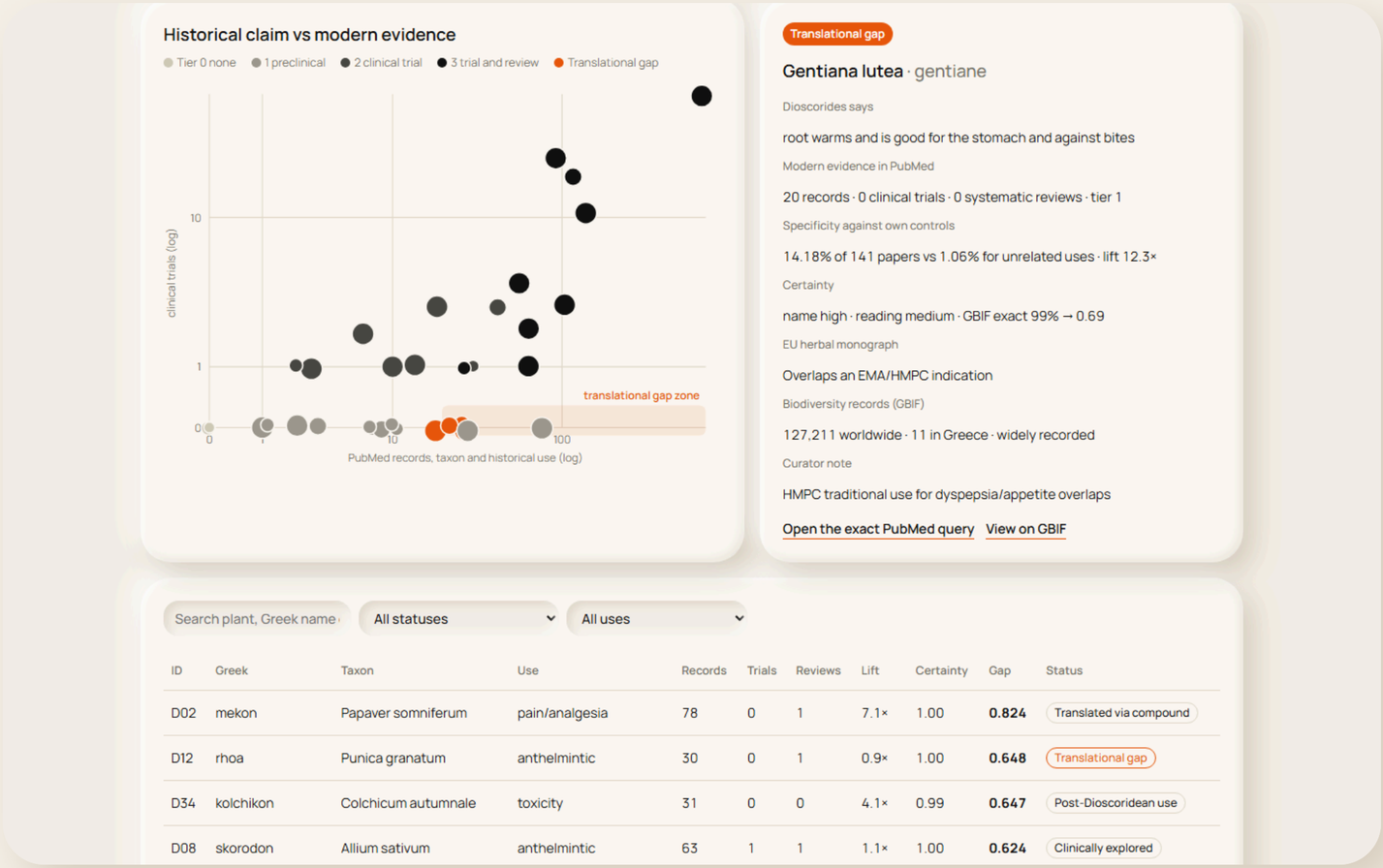
ITEA

Willow → salicin → aspirin

KOLCHIKON

Autumn crocus → colchicine

An atlas you can audit claim by claim



What it does not claim

- Co-occurrence measures attention, not efficacy
- Trial counts say nothing about trial results
- The seed table needs checking against the Wellmann edition
- GBIF records measure sampling, not abundance
- A public domain text: no living community knowledge was taken. Extending to it needs consent and Nagoya benefit sharing
- Toxic taxa flagged. Not medical advice

A method for every historical pharmacopoeia

Drug discovery

A ranked, auditable shortlist of historically grounded hypotheses for clinical study.

History of medicine

A quantitative view of which ancient knowledge survived into modern pharmacology.

Conservation

Demand signals tied to occurrence data, from lost silphion to Greek endemics.

Next: Greek NLP over all ~800 chapters · Byzantine, Arabic and Latin name mapping · trial outcome grading from ClinicalTrials.gov · compound links via LOTUS and Wikidata · IUCN risk

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